MILITARY SPECIFICATION

DIATRIZOATE SODIUM FOR ORAL SOLUTION, 250 Gm

1. SCOPE

1.1 This specification covers a water soluble powder of Sodium Diatrizoate, suitable for preparing radiopaque solutions.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-665	Boxes, Paperboard, Metal Stayed.
PPP-B-676	Boxes, Set-Up, Paperboard.
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the J.B. Lippincott Company, Philadelphia, Pa.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
General Regulations for its Enforcement.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D. C. 20204.)

U. S. PHARMACOPOEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa.)

3. REQUIREMENTS

3.1 Material. Shall be a water-soluble powder of sodium diatrizoate (sodium 3,5-diacetamido-2,4,6-triiodobenzoate) containing 0.1 percent polysorbate 80 as a dispersing agent and caramel as a coloring agent.

3.1.1 Assay. The powder shall assay to contain not less than 98.0 percent and not more than 102.0 percent of sodium diatrizoate, calculated on the dry basis, when assayed as specified in 4.3.2.1.

3.1.2 Identification. The powder shall respond to identification test A for sodium diatrizoate ingredient, described in 4.3.1.1.1.

3.1.3 Moisture. Shall contain not more than 10 percent moisture when determined by the U.S.P. Karl Fischer method.

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3.1.4 Heavy metals. The heavy metals content shall be not more than 20 parts per million (as Pb), when determined as specified in 4.3.2.2.

3.1.5 Clarity and color. Shall be a clear brownish-yellow solution having a color value range of 225 to 235 when tested as described in 4.3.2.3.

3.2 Ingredients.

3.2.1 Sodium diatrizoate. The sodium diatrizoate shall comply with the following requirements:

3.2.1.1 Description. Shall be an odorless white powder, soluble in water, slightly soluble in alcohol, and practically insoluble in acetone and ether.

3.2.1.2 Identification. The sodium diatrizoate shall comply with the tests specified in 4.3.1.1.1.

3.2.1.3 pH (50 percent solution). The pH of a 50 percent solution shall be between 7.0 and 9.5, when determined as specified in 4.3.1.1.2.

3.2.1.4 Moisture. Shall contain not more than 10 percent moisture when tested as specified in 4.3.1.1.3.

3.2.1.5 Heavy metals. The heavy metals content shall be not more than 20 parts per million (Pb) when tested as described in 4.3.1.1.4.

3.2.1.6 Assay. Shall assay to contain not less than 98.0 percent and not more than 102.0 percent sodium diatrizoate when tested as specified in 4.3.1.1.5.

3.2.2 Polysorbate 80. The Polysorbate 80 shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplement or revisions thereto.

3.2.3 Caramel color (dry). Shall be a dark brown powder prepared from caramel by removing substantially all the residual or uncarmelized sugars. The modified caramel shall comply with the following requirements:

3.2.3.1 Caramel. The caramel used in preparing the caramel color shall be of N.F. quality.

3.2.3.2 Color. The caramel color (dry) shall comply with the test for color as specified in 4.3.1.3.1.

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3.2.3.3 Moisture. The moisture content shall not exceed 5 percent when dried at 70° C. for 4 hours.

3.2.3.4 pH. The pH of a 10 percent w/v solution shall be between 2.8 and 3.4.

3.3 Workmanship. Workmanship shall be first class throughout. The material and its container shall be free from any defects which detract from their appearance or impair their serviceability.

3.4 Delivery schedule. Not more than 6 months shall have elapsed from the date of manufacture of the product, to the date of delivery to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Lot. For purpose of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of finished powder produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.2 Sampling. Sampling shall be conducted in accordance with the procedures set forth in MIL-STD-105, with an Acceptable Quality Level (AQL) of 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

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4.3 Tests.

4.3.1 Ingredients.

4.3.1.1 Sodium diatrizoate.

4.3.1.1.1 Identification. Identification shall be determined as follows:

A. Heat approximately 500 mg of sodium diatrizoate in a porcelain dish over a flame; the residue shall decompose and liberate iodine vapors.

B. Prepare a solution of sodium diatrizoate in alcohol to contain in each 100 ml, 0.5 mg of sodium diatrizoate; the ultra-violet absorption spectrum of this solution shall exhibit inflections at the same wavelengths as that of U.S.P. Sodium Diatrizoate Reference Standard, similarly measured.

C. The infra red absorbtion spectrum of a potassium bromide dispersion of the Sodium Diatrizoate shall exhibit maxima only at the same wavelengths as a similar preparation of U.S.P. Sodium Diatrizoate Reference Standard. (See figure 1 which is provided for informational purposes.)

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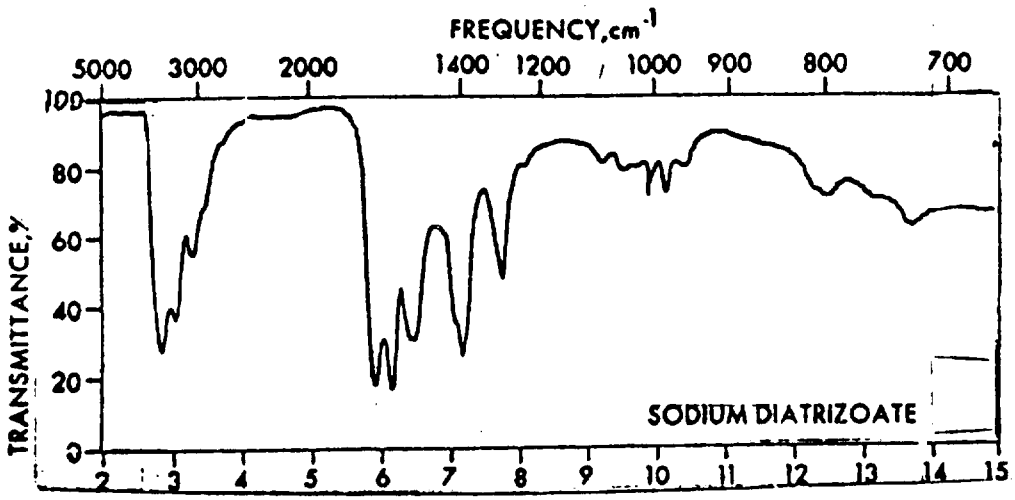


figure 1

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4.3.1.1.2 pH. Determine pH of a 50 percent w/v solution of sodium diatrizoate electrometrically using glass-calomel electrodes.

4.3.1.1.3 Moisture. Determine the moisture content by the U.S.P. Karl Fischer method (U.S.P. XVI, page 939), modified as follows:

> Add about 25 ml of methanol to the titration beaker and titrate to the endpoint with Karl Fischer Reagent. Transfer 500 mg of sample to the titration beaker, stir vigorously and titrate with Karl Fischer Reagent to the endpoint.

4.3.1.1.4 Heavy metals. Conduct test for heavy metals, as follows:

Reagents:

a. Sodium Sulfide, T.S. (U.S.P. XVI, page 1076).

> Dissolve 10 Gm of $Na_2S.9H_2O$ in water distilled to make 100 ml. Filter through a small plug of glass wool. The solution shall be clear and colorless.

b. Lead Nitrate Stock Solution (U.S.P. XVI, page 899).

> Dissolve 159.8 mg of lead nitrate in 1000 ml of distilled water which contains 1.0 ml of nitric acid. Prepare and store in glass containers free from soluble lead salts.

c. Standard Lead Solution.

> Dilute 10 ml of the lead nitrate stock solution with distilled water to 100 ml. This solution must be freshly prepared. When 2.0 ml of the standard lead solution is employed to prepare the standard to be compared with a solution of 1 Gm of sodium diatrizoate, the comparison solution. Thus prepared contains the equivalent of 20 parts of lead per million parts of sample tested. Procedure: Transfer 2 ml, containing 1 Gm of Sodium Diatrizoate, to a flat bottom tube, add 10 ml of sodium sulfide T.S. and mix contents of tube. Compare with a standard containing 20 p.p.m. lead (2.0 ml) and 10 ml of sodium sulfide T.S. in a matched flat bottom tube.

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4.3.1.1.5 Assay. Conduct test for sodium diatrizoate, as follows: (Method from organic iodine by reduction with zinc.)

Reagents:

5 percent w/v solution of sodium hydroxide.

Granular zinc reagent grade, 20-30 mesh.

Glacial acetic acid.

Tetrabromophenolphthalein ethyl ester 0.1 percent glacial acetic acid.

0.05N $AgNO_3$.

Procedure: Accurately weigh 0.300 Gm of sample into a 250 ml glass-stoppered Erlenmeyer flask. Add 30 ml of 5 percent sodium hydroxide and 0.5 Gm of granular zinc. Attach the flask to a reflux condenser and boil the mixture gently for one (1) hour. Allow to cool. Wash the condenser with 20 ml of distilled water. Filter the contents of the flask through a 150-ml fritted glass Buchner funnel of medium porosity and collect filtrate in a 250-ml suction flask. Wash funnel with small portions of distilled water. Acidify filtrate with 15 ml of glacial acetic acid which will adjust the pH to approximately 4. Add 1 ml of tetrabromophenolphthalein ethyl ester indicator and titrate with 0.05N silver nitrate. The endpoint is reached when the color of the precipitate changes from yellow to green.

4.3.1.2 Polysorbate 80. The polysorbate 80, used in the manufacture of the finished powder shall be tested by the methods described in the U.S.P.

4.3.1.3 Caramel color.

4.3.1.3.1 Color. Determine the absorbancy of a 0.1 percent aqueous solution of caramel at 520 mu in a Beckman spectrophotometer, using distilled water as a blank. The solution absorbancy less the blank absorbancy is in the range of 0.248 to 0.277.

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4.3.2 Finished product.

4.3.2.1 Assay. Determine the percent sodium diatrizoate in the finished product employing the method specified in 4.3.1.1.5.

4.3.2.2 Heavy metals. Carefully ignite 1.0 Gm of sample with sulfuric acid in a platinum dish. Take up the residue in 23 ml of distilled water and 2 ml of diluted acetic acid. Determine the heavy metals content by the U.S.P. method.

4.3.2.3 Clarity and color. Dissolve 0.5 Gm of finished powder in 5 ml of distilled water. A clear, brownish-yellow solution shall result. Dilute with an equal volume of distilled water and determine the color value (absorbancy x 1000) at 420 mu.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified herein and shall be in accordance with all the applicable requirements of PPP-C-186.

5.1.1 Immediate containers. Each immediate container shall comply with the following classification:

GROUP D CLASS 1 - Parchment lined.

Shall have a friction lid fitted with a suitable, metal, lever-type (lift) opener.

A plastic spoon capable of holding one tablespoonful (approximately 10 Gm) of the powder shall be packaged within each can.

5.2 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act, and shall include the information required below:

5.2.1 Immediate containers. Each immediate container label shall bear the following information. However, the information is not required to appear in the sequence indicated.

(a) the item identification designated as "DIATRIZOATE SODIUM FOR ORAL SOLUTION"

(b) the quantity of contents designated as "250 Gm"

(c) the stock number

(See additional labeling on page 10)

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(d) the lot or control number

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designation shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) indication that the product:

1. contains the dispersing agent polysorbate 80, 0.1%.
2. is colored with caramel

(g) The following statements or similar statements:

1. Warning: NOT for Parenteral Injection.

2. Caution: Federal law prohibits dispensing without prescription.

3. For GASTROENTERIC RADIOGRAPHY.

4. Read accompanying circular.

(h) table for preparing solutions.

Solution (Approximate) (%)	Level measuring spoon* of powder (per 100 ml diluent)
10	1
15	1-1/2
20	2
25	2-1/2
40	4

*Enclosed measuring spoon holds approximately 10 Gm.

(i) the statement in juxtaposition to the table: "Solutions may be sweetened or flavored (for example; with vanilla, lemon, chocolate); the diluent may be water, milk or a carbonated drink."

(j) dosage information (oral and as an enema).

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5.2.2 A circular, brochure, or other printed matter shall accompany each immediate container setting forth as a minimum: Indications for use, dosage and administration, precautions and contraindications, and side effects.

5.3 Packaging.

5.3.1 Unit of issue. One (1) can, as specified constitutes one unit of issue.

5.3.1.1 Intermediate package. The number of units of issue specified in table I, column 2, for Procedure Code No. 18, indicated in column 1, shall be packaged in an intermediate package constructed in accordance with the applicable paragraph referenced in column 3. Commercial colors shall be acceptable when carton conforming to PPP-B-566, PPP-B-665, or PPP-B-676 is utilized.

5.4 Packing. The number of units contained in the intermediate packages and total number of units of issue for the applicable procedure code, indicated in column 4, shall be overpacked in an exterior container, constructed in accordance with the applicable paragraph referenced in column 5 (level B), or column 6 (level A), for the level of protection specified in the procurement document. Bursting strength of carton shall be in accordance with special requirements in table I of PPP-B-636, in lieu of bursting strength specified in the applicable paragraph referenced in column 5 (level B). Method I closure of PPP-B-636 shall be utilized on level B packs. In addition, in line 10 of 5.4.4, delete "IV" and substitute "III; or PPP-B-636, type CF, class weather-resistant; at end of 5.4.4.1, add "Case liner shall not be required for fiberboard boxes." Add the following new paragraph: "5.4.4.1.1 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of PPP-B-636. In addition, waterproofing shall conform with 30.4 of PPP-B-636."

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5.5 Marking.

5.5.1 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number, and name of contractor shall be shown.

5.5.2 Exterior containers. Exterior container shall be marked in accordance with MIL-STD-129. The terms "Stock Number" or "FSN" shall not be indicated. Lot (control) numbers shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.4).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION MILITARY SPECIFICATION DIATRIZOATE SODIUM FOR ORAL SOLUTION, 250 Gm MIL-D-36379(DM) dated, 9 February 1965

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1426

SND PPSO 12502

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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BROOKLYN, N.Y., 11232

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OFFICIAL BUSINESS

DEFENSE SUPPLY AGENCY

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